My use of AI - Livesport

My workflow during the task was iterative. I usually started with my own idea of the validation logic, then used AI tools to speed up repetitive SQL generation or SQL syntax debug or suggest additional checks. After that, I manually reviewed the outputs, adjusted queries, and interpreted the results myself. Additionally, for my own interest I asked AI for interpretation, which mostly supported my conclusions.

I validated both my own and AI-generated SQL queries by comparing row counts (in .xls), checking join cardinality, and manually inspecting sampled records (in .xls). For example, when joins produced unexpectedly high mismatch counts, I verified whether duplicates in the datasets were inflating the results before drawing conclusions.

During the task, I used AI tools ChatGPT and Google Gemini for

- generating interpretations of results
- generating repetitive SQL queries
- shortening and improving comments in the SQL document
- fixing syntax errors
- double-checking the PDF report for English grammar and readability
- brainstorming additional validation ideas and edge cases and brainstorming following steps
- validating whether my approach was logically consistent

Where AI failed in this task

- AI did not correctly understand how to approach duplicate detection in some cases.
- When checking missing values, AI suggested accounting for empty strings as well, but some column data types did not support this approach, which caused SQL errors so I had to manually adjust querries.
- Some AI-generated interpretations of the results were misleading or too confident without enough evidence from the data itself.

AI was especially useful for repetitive query generation and brainstorming validation scenarios, which allowed me to spend more time interpreting anomalies and reasoning about possible pipeline issues.

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Prompts:

- Generate a DuckDB query that checks NULL values for every column in a table using COUNT(*) FILTER.
- Generate SQL queries comparing DISTINCT dimension values between two tables using EXCEPT.
- Suggest additional data quality checks for datasets containing user activity, revenue metrics, signup dates, and tracker mappings.
- Explain why a JOIN on user_id and as_of may produce inflated mismatch counts.

Overall, I see AI as a productivity multiplier, useful tool when handled correctly, rather than a replacement for analytical thinking. In data engineering tasks, validation is critical, so AI outputs should always be verified against the underlying data and logic.